➢ *ORAL PRESENTATION*

Predicting Maternal Health Risks with Advanced Machine Learning Models

Celal Oğuz Kurtoğlu[1*] (ORCID: 0009-0008-2937-4459), Serkan Kartal[1](ORCID: 0000-0001-9801-8986)

[1]Cukurova University, Engineering Faculty, Computer Engineering Department, Adana, Turkey.

*Corresponding author e-mail: celaloguz.kurtoglu@gmail.com

Abstract
In this study, the performance of advanced machine learning and neural network algorithms was examined to predict the level of maternal health risk based on data obtained from pregnant women. Pregnancy carries potential risks for both the mother and the fetus, and early detection of these risks is crucial for timely and effective interventions. Various neural network architectures and machine learning models, including XGBClassifier, RandomForestClassifier, LGBMClassifier, and Logistic Regression (L1), were utilized to perform this task, and their performances were assessed with the well-known metrics such as precision, recall, and F1 score. The selection of these models is based on their proven success in prediction tasks and their widespread use in similar prediction processes in the literature. To improve the prediction performance, the hyperparameters of the Models are optimized using FLAML (Fast and Lightweight AutoML), an automatic hyperparameter tuning tool. The results demonstrate that Gradient Boosting-based algorithms outperform other models in predicting maternal health outcomes.

Keywords: Machine Learning, Neural Network, Hyperparameter Optimization, Maternal Health

INTRODUCTION

Maternal health is a critical aspect of healthcare, vital for the well-being of women during pregnancy, childbirth, and the postpartum period. High maternal mortality rates, particularly in developing countries, highlight the severity of challenges in this field (WHO, UNICEF, UNFPA, and The World Bank, 2012). Ensuring the health of both mother and fetus requires effective interventions and treatments. Artificial intelligence (AI) is set to significantly transform medicine, with the potential to enhance the experiences of both healthcare providers and patients (Rajpurkar et al., 2022). Through sophisticated data analysis, AI methodologies offer the capability to distinguish and anticipate various health complications of maternal well-being. This analytical ability equips healthcare practitioners with invaluable insights into the health status of expectant mothers, facilitating prompt diagnosis and interventions aimed at preventing adverse outcomes.

The development of robust AI models, adept at accurately forecasting health risks during pregnancy, assumes paramount importance in the realm of maternal and infant healthcare preservation and enhancement (Togunwa et al., 2023). A novel machine learning (ML) approach tailored for pregnancy and childbirth achieved an accuracy of 70.21%, demonstrating its reliability over traditional methods such as decision tree (DT) and random forest (RF) (Pawar et al., 2022). Similarly, Nirmala and Kambli (2023) found that a RF model achieved the highest accuracy rate of 83% in predicting maternal risk levels, surpassing Neural Network (NN) (57%) and kNN (53%) models. These AI algorithms utilized features such as age, systolic blood pressure, diastolic blood pressure, blood sugar levels, body temperature, and heart rate risk level predictions. The findings of this research underscore the effectiveness of RF models in accurately assessing maternal health risks.

Integrating AI into maternal health not only enhances predictive capabilities but also supports healthcare professionals in making informed decisions. By analyzing large datasets encompassing diverse health metrics, AI systems can identify subtle patterns and correlations that might be overlooked by human clinicians. This capability is pivotal in providing personalized healthcare and improving maternal outcomes globally. In this study, various ML and deep learning algorithms were used to determine the levels of risk in maternal health. However,

determining the necessary hyperparameters for these algorithms to perform at their best is a complex process. Therefore, the Fast and Lightweight AutoML (FLAML) library was used to automatically adjust hyperparameters, and the abilities of these algorithms to determine levels of risk in maternal health were evaluated.

MATERIALS and METHODS

Dataset

The dataset was sourced from the UCI Machine Learning Repository. The given data were collected from hospitals, maternity health services, and community clinics in rural areas of Bangladesh. During the data collection process, an IoT-based advanced risk monitoring system was employed to effectively track the health conditions of pregnant women (Ahmed and Kashem, 2020).

The dataset includes several significant features that are critical in assessing the risk factors associated with maternal mortality. These features are:

- Age: Whole number indicating the woman's age in years.
- SystolicBP: Integer showing the upper blood pressure value in millimeters of mercury (mmHg).
- DiastolicBP: Integer value representing the lower value of blood pressure measured in millimeters of mercury (mmHg).
- BS (Blood Sugar): Integer indicating the lower blood pressure value in millimeters of mercury (mmHg).
- BodyTemp (Body Temperature): Integer value representing body temperature in degrees Fahrenheit (F).
- HeartRate: Integer representing the normal resting heart rate in beats per minute (bpm).
- RiskLevel: Categorical target variable indicating the predicted risk intensity level during pregnancy based on the aforementioned attributes. The distribution of the estimated Risk Level values is shown in Fig. 1.

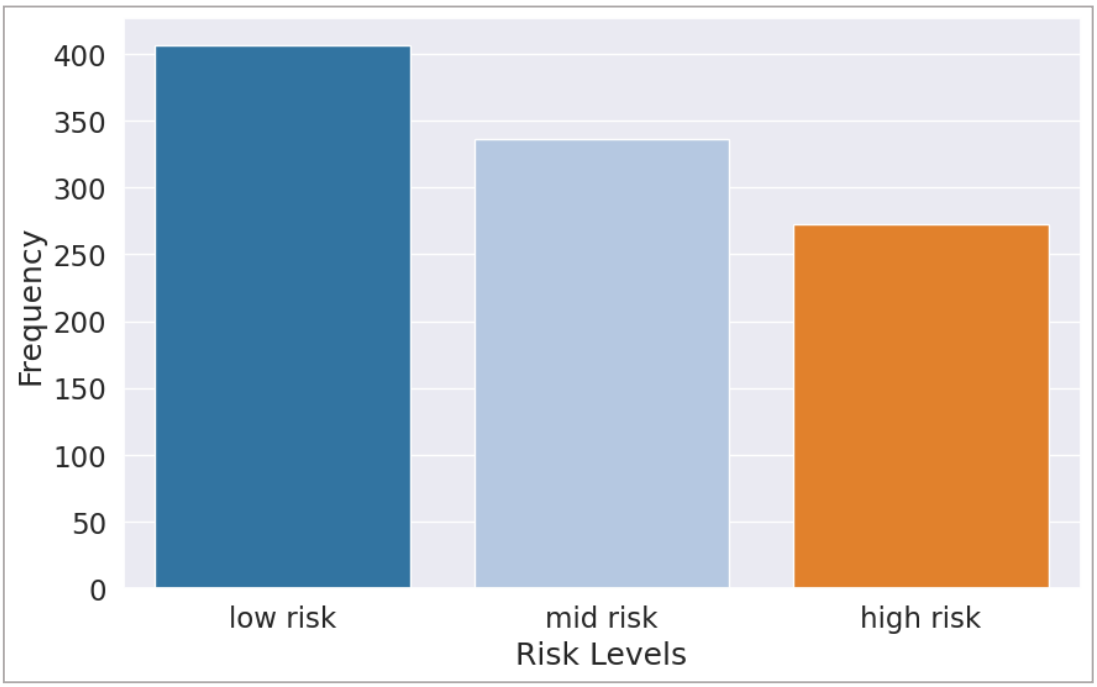

Fig. 1. Distribution of Risk Levels.

The distribution and density of data are illustrated in the violin plot given in Fig 2. This graphical representation facilitates the comparison of density and distribution across different variables. It aids in determining the central tendency and extreme values of the data, while clearly indicating the specific value ranges where the data is concentrated.

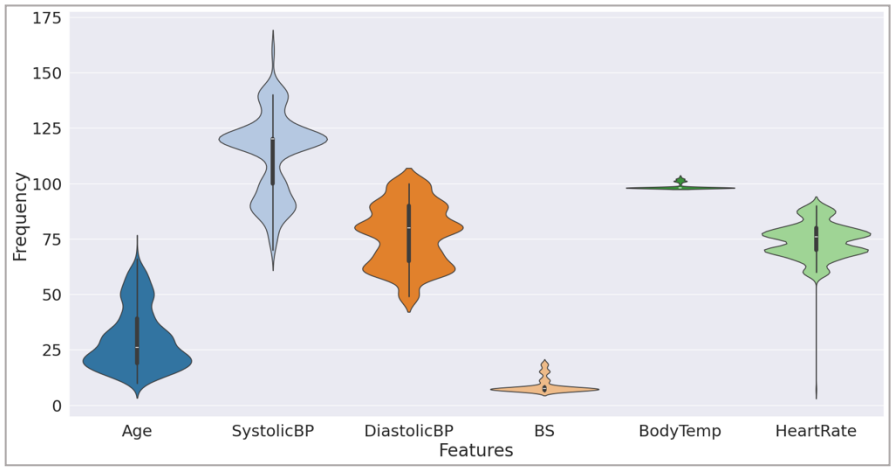

Fig. 2. Visualization of the distribution and density of data.

A review of this figure indicates that there are no significant imbalances that could impact the results of the study. Following this assessment, the dataset was subjected to preprocessing steps, including Ordinal Encoding, to ensure it was suitable for ML models. After encoding, the dataset was split into 80% for training and 20% for testing.

Machine Learning & Deep Learning Models

In this study five different ML models were utilized for maternal health risk prediction. Each model was chosen for its unique strengths and effectiveness in prediction tasks. A brief explanation of each model is given below.

Random Forest

The first method employed for risk prediction was the RF algorithm (Breiman, 2001). RF is a powerful ensemble learning technique that consist of multiple DT. Each tree is built using a randomly selected subset of the data, which helps to enhance the model's generalization ability. One major benefit of random forests is their ability to mitigate overfitting. This is achieved by aggregating predictions from numerous decision trees trained on varied subsets of data, resulting in robust and broadly applicable outcomes.

XGBoost

XGBoost, short for "Extreme Gradient Boosting" is another highly effective ML algorithm (Chen & Guestrin, 2016). XGBoost builds on the principles of gradient boosting, which combines the predictions of several base learners to improve accuracy. It is particularly noted for its performance and speed in classification and regression tasks, making it a popular choice for structured data analysis.

LightGBM

LightGBM (Light Gradient Boosting Machine) is a gradient boosting framework that uses a histogram-based approach to discretize continuous variables into discrete bins, significantly reducing computational costs (Ke et al., 2017). This optimization allows for faster training times and efficient resource utilization. LightGBM is well-suited for handling large datasets with numerous features.

L1 Regularized Logistic Regression

L1 regularized logistic regression, commonly known as Lasso regression, is extensively used in classification tasks with many features (Lee et al., 2006). This method addresses the convex optimization problem by adding a penalty term to the loss function, which encourages sparsity in the model coefficients. However, it faces scalability challenges with large datasets when using traditional algorithms.

Neural Networks

NNs have also been employed for maternal health risk prediction (LeCun et al., 2015). These models are capable of capturing complex, non-linear relationships within the data, making them suitable for tasks involving high-dimensional features. NNs consist of multiple layers of interconnected nodes (neurons) that can learn hierarchical representations of the input data, showing remarkable performance in various predictive tasks across healthcare domains.

Hyperparameter Optimization

Hyperparameter optimization involves finding the best set of hyperparameters for an ML model. Hyperparameters are configuration settings used to tune the behavior of the learning algorithm, such as learning rate, regularization strength, and the number of hidden layers in an NN. Unlike model parameters that are learned from the training data, hyperparameters are set before the training process begins. These parameters can significantly impact the model's performance (Yang and Shami, 2020). However, choosing these parameters is not straightforward and often involves trial and error. While Grid Search and Random Search are commonly employed for hyperparameter optimization, this study utilized the FLAML library developed by Microsoft (Wang et al., 2021). Grid Search and Random Search are computationally intensive and time-consuming, whereas FLAML offers a more efficient and lightweight approach. By imposing a time limit, FLAML identifies suitable parameters for specified algorithms, yielding optimal results within the given constraints.

RESULTS and DISCUSSION

FLAML was used to automate the hyperparameter tuning process. Each model was run for one hour to find the best configuration. After the training process completed, models performance evaluated on the test dataset. For the performance evaluation, widely used performance metrics named precision, recall, accuracy and F1 score were calculated. The performance of each model on the test dataset are given below (Fig. 3).

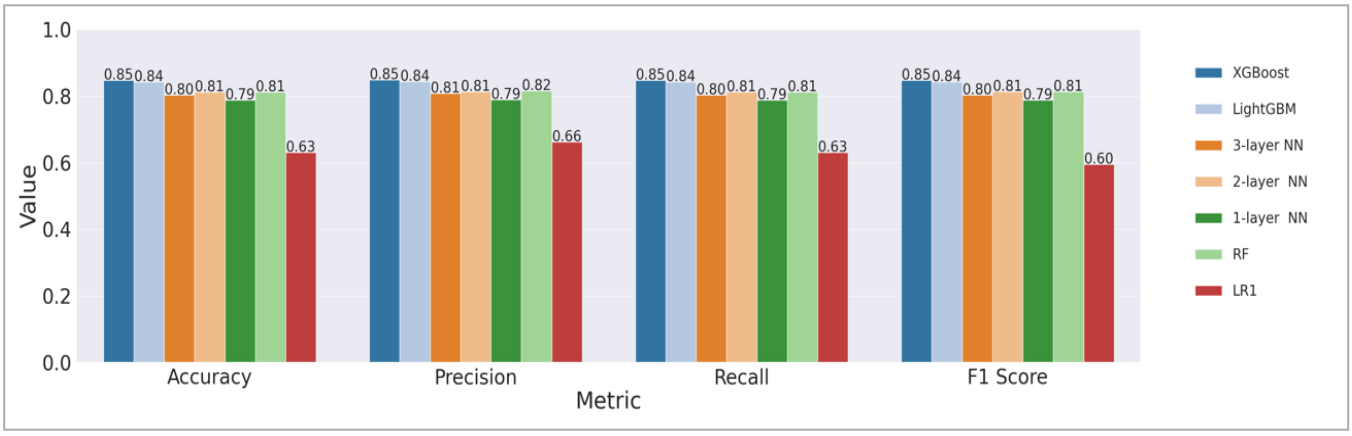

Fig. 3. Model Performance Metrics with FLAML Optimization.

In evaluating the dataset using various ML models, L1 Regularization (LR1) serves as the baseline, achieving an accuracy of 0.630542 and an F1 score of 0.595523. LR1's performance highlights its limitations in capturing complex data relationships. Moving to NNs, a one-layer model achieves an accuracy of 0.788177 and an F1 score of 0.788304, surpassing LR1 but falling short of more intricate architectures. Introducing a second layer shows marginal improvement, aligning closely with RF in accuracy and precision. However, deeper architectures like a three-layer NN exhibit a slight decline in metrics, hinting at potential overfitting or training inefficiencies due to weight vanishing. Among tree-based algorithms, RF demonstrates robust performance with an accuracy of 0.812808 and an F1 score of 0.813163, highlighting its effectiveness in handling complex datasets. Nevertheless, it is outperformed by XGBoost, leading with an accuracy of 0.847291 and an F1 score of 0.847286, closely followed by LightGBM with an accuracy of 0.842365 and an F1 score of 0.842617. These two boosting models surpass NNs and Bagging models in all evaluation metrics, showcasing their superior ability in maternal health risks prediction. In summary, XGBoost stands out as the best performer, attaining the highest accuracy, precision, recall, and F1 score compared to all other models assessed. Its robust performance across all metrics makes XGBoost the optimal choice for predictive modeling in this dataset, ensuring reliable performance in similar contexts.

CONCLUSION

This study evaluated the effectiveness of various ML models and NN architectures in predicting maternal health risks based on pregnant women data. The primary objective was to facilitate early detection and intervention for potential maternal health issues by identifying the most accurate predictive models. The results indicated that Gradient Boosting algorithms, specifically XGBoost and LightGBM, outperformed other models in terms of accuracy, precision, recall, and F1 score. Random Forest and a two-layer NN also showed strong performance but were slightly less effective than the Gradient Boosting models. L1 Regularization was the least effective in capturing the complexities of the dataset. This study highlights the potential of advanced ML techniques, particularly those utilizing automated hyperparameter tuning, in improving the prediction of maternal health risks, thereby supporting timely and effective healthcare interventions.

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